

Reading Relations, Loops, Flux, and Condensates from the Closure of Complex Numbers — A Minimal-Assumption Thought Experiment That Shaves Off Axiom Candidates

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Abstract

This paper proposes no new algebraic law and no new equation of motion. The starting point is only the ordinary complex number $z \in \mathbb{C}$, and the only facts used are the known ones that complex numbers are closed under addition, multiplication, and squaring. In particular, for any finite complex sequence $\{z_n\}_{n=1}^N$ we have $z_n^2 \in \mathbb{C}$, so setting $Z_n := z_n^2$ we can read

$$\sum_{n=1}^N z_n^2 = \sum_{n=1}^N Z_n = K, \quad K \in \mathbb{C}.$$

Furthermore, any $K \in \mathbb{C}$ can be written as $K = C^2$ for some $C \in \mathbb{C}$, so

$$\sum_n z_n^2 = C^2$$

is not a new physical axiom but a notation that stays within the closure of the complex numbers.

The subject of this paper is not this trivial fact itself but its **order of readout**. In the thought experiment, we questioned, one by one, complete networks, external standards, ratios, equality tests, zero closure, and the presupposition of triangles and vertices, removing hidden assumptions against namelessness. As a result, when the squared quantities Z_n are read as ordinary complex relations, we found that sums of multiple relations, loops, loop residuals, face fluxes, cancellation of shared parts, closed polyhedra as condensates, and further the hierarchization of condensates can all be described while staying within the same complex-number type.

If the sum of three relations is zero it can be read as a loop; if nonzero it carries a residual K_f . When multiple loops are glued together with orientations, shared edges cancel with opposite signs, and only the quantities remaining on the external boundary survive as effective flux. Since the face residual K_f is itself a complex number, it can be reused as a relation at the next level of the hierarchy. This hierarchy has the characteristic that edge, face, and cell types need not be postulated separately: the recursion runs inside the same $\mathbb{C}$.

At the end, the paper compares structural similarities with lattice gauge theory, coarse-graining of spin networks, closure defects, the Bianchi identity, and bubble networks. However, none of these is placed at the starting point. The candidate novelty of this paper is not a simplification of existing theories but **the fact that, starting only from the trivial closure of ordinary complex numbers and shaving off hidden assumptions along the way, we obtain a minimal interpretive framework that reads off loops, curvature, flux, and condensates.**

Keywords: complex numbers, squared quantities, loop, closure, closure defect, flux, curvature, condensate, namelessness, minimal assumptions, lattice gauge theory

1. What This Paper Does and Does Not Claim

1.1 What is not claimed

This paper does not claim the following as new mathematical theorems.

1. That the square of a complex number is a complex number.
2. That a finite sum of complex numbers is a complex number.
3. That every complex number has a complex square root.
4. That if a sum of complex vectors is zero it can be drawn as a closed polygon.
5. That on an oriented boundary a shared element appears with opposite signs.

These are known mathematics.

1.2 The subject of this paper

What this paper addresses is **in what order, and without which additional assumptions, the known facts above can be read as physical structure.**

Therefore this paper is not a “paper proposing a new equation” but

a minimal-assumption readout paper on known identity structures.

1.3 Methodology

At each stage of the thought experiment, before introducing a new structure we ask:

- Is the structure derived from the algebra of complex numbers?
- Are we implicitly importing an external standard?
- Does it require names of individuals or vertices?
- Are we granting in advance a complete graph, distance, dimension, coordinates, or time?
- Are we placing a “same/different” detector as a primitive operation?
- Are we importing an existing physical theory back into the foundations?

Only what cannot be derived yet is needed remains as an **axiom candidate**; whatever can be derived is removed from the axiom candidates.

1.4 Revision note for v1.2: a hidden assumption about the completeness of many-body relations

After completing v1.1 of this paper, we read Anegawa, Miyata, Suzuki, Tamaoka (2026) [6]. Although that paper works in a different framework — quantum information and holography — it exhibits examples in which, for a fixed tripartition, quantities measuring genuine multi-partite entanglement can continue to evolve in time even after the ordinary entanglement entropies of the subsystems have saturated.

Prompted by examining that paper, we noticed that v1.1 of the present paper implicitly retained the assumption

$\{Z_e, K_f\}$ suffices to exhaust the effective relational information of a many-body system.

That is, we had effectively presupposed that the pairwise complex relations Z_e , together with the set of loop residuals K_f constructed from them, suffice to read the relational structure of a many-body system. This completeness, however, has not been derived from the complex-number algebra of this paper.

The results of Anegawa et al. [6] neither directly support nor refute the present model, and K_f cannot be identified with multi-entropy. On the other hand, the concrete example that **irreducible many-body relations can remain even after low-order readouts saturate** shows that the above completeness cannot be taken for granted.

Therefore, in v1.2 this assumption is removed from the derived properties, and its validity is stated explicitly as a problem for future work. This revision is the result of applying to the paper itself the very methodology the paper adopts: “when a hidden assumption is discovered, delete it or put it on hold.”

2. Starting Point: A Single Complex Number

Take as the starting point

$$z \in \mathbb{C}.$$

Here we assume no particles, edges, vertices, faces, space, time, gauge group, or network.

From the closure of the complex field,

$$z^2 \in \mathbb{C}.$$

This is utterly trivial. But in this paper we pursue how much structure can be read without breaking this triviality.

Even when a single complex number is multiplied by itself,

$$z \mapsto z^2,$$

the result is again a single complex number. Therefore, at this stage one cannot say “one body has split into two” or “a new vertex has appeared.” The algebraic type remains

$$\mathbb{C} \rightarrow \mathbb{C}.$$

What matters here is that **the generation history and the number of individuals cannot be recovered from the results of complex operations alone**. Hence, to discuss structure, the case in which multiple complex relations appear simultaneously must be considered separately.

3. Two Complex Numbers: Take the Product, Not a Comparator

Suppose there are two complex numbers

$$z_1, z_2 \in \mathbb{C}.$$

One might first consider taking the ratio z_2/z_1 as the relational quantity. However, reading a ratio as “larger/smaller” or “same/different” risks smuggling in a comparison standard. Likewise, placing the equality test $z_1 = z_2$ as a primitive operation is too strong for the purposes of this paper.

The product

$$z_1 z_2,$$

on the other hand, requires no external standard. From the closure of complex numbers,

$$z_1 z_2 \in \mathbb{C}.$$

Expanding into real and imaginary parts, for

$$z_1 = a + ib, \quad z_2 = c + id$$

we have

$$z_1 z_2 = (ac - bd) + i(ad + bc).$$

The result is again a single complex number.

Using the same value twice gives

$$z z = z^2,$$

so the self-product is a special case of the two-body product. But there is no need to detect from outside that the values are equal and split the structure into cases. Algebraically, one simply takes the product.

Also, since the complex numbers have no zero divisors,

$$z_1 z_2 = 0$$

implies that at least one factor is zero. Hence the minimal algebraic boundary that appears without any external scale is “zero / nonzero.” In this paper, however, we do not elevate this to a new physical axiom.

4. Reading the Squared Quantity as the Basic Quantity

Here we perform the main rereading of this paper.

For each complex number define

$$\boxed{Z_n := z_n^2}.$$

Since the square of a complex number is a complex number,

$$Z_n \in \mathbb{C}.$$

Then the expression

$$\sum_n z_n^2,$$

which often looks like a special form, is simply

$$\boxed{\sum_n Z_n},$$

an ordinary sum of complex numbers.

Therefore

$$\sum_n z_n^2 = C^2$$

is also, once written as

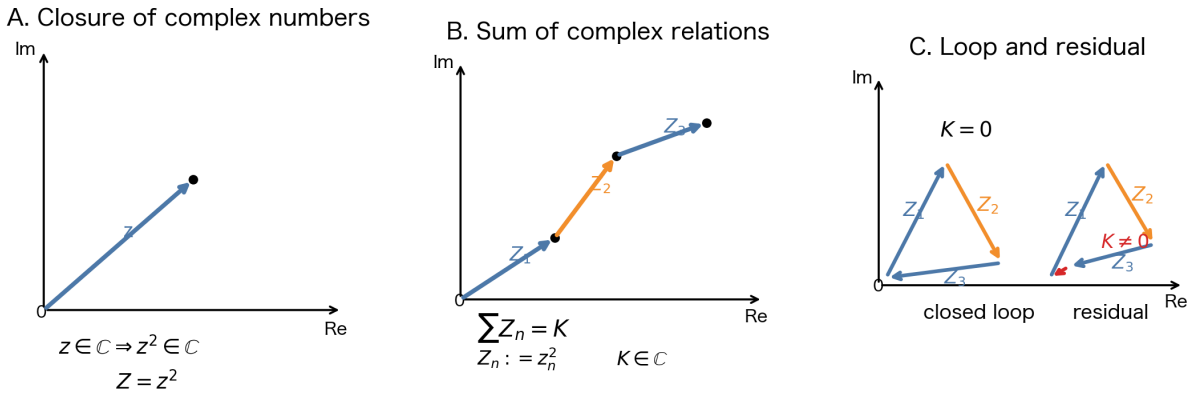
$$\boxed{\sum_n Z_n = K}, \quad K := C^2 \in \mathbb{C},$$

just an ordinary complex sum.

At this point there is no need to treat C^2 as “special because it is a square.” In the space of squared quantities, $K = C^2$ is an ordinary complex number.

This rereading is the core of this paper.

Fig. 1 From the closure of complex numbers to the loop residual



The trivial identity is read as a sum of complex relations and a loop residual.

Figure 1: Fig. 1 From the closure of complex numbers to the loop residual

Fig. 1 From the closure of complex numbers to the loop residual. The trivial identity $\sum z_n^2 = C^2$ is reread as the sum $\sum Z_n = K$ of complex relations $Z_n := z_n^2$. $K = 0$ represents closure; $K \neq 0$ represents a loop residual.

5. An Auxiliary Two-Body Thought Experiment: When the Sum Is Real

Consider an auxiliary thought experiment in which, for two complex numbers, the only constraint imposed is

$$z_1^2 + z_2^2 = R^2 \in \mathbb{R}.$$

Writing the squared quantities as

$$Z_1 = z_1^2, \quad Z_2 = z_2^2,$$

this is simply

$$\boxed{Z_1 + Z_2 = R^2}.$$

Subtracting the center,

$$W_1 := Z_1 - \frac{R^2}{2}, \quad W_2 := Z_2 - \frac{R^2}{2},$$

we get

$$\boxed{W_1 + W_2 = 0},$$

that is,

$$W_2 = -W_1.$$

In the original z representation this involves square-root branches, but once the squared quantity Z is taken as basic, the two bodies simplify to an antisymmetric configuration about the center $R^2/2$ on the real axis.

This auxiliary example shows the advantage of “reading squared quantities as basic quantities,” but the paper does not require $R^2 \in \mathbb{R}$ throughout.

6. Three Bodies: The First Nontrivial Loop

For three squared quantities, let

$$Z_1 + Z_2 + Z_3 = R^2 \in \mathbb{R},$$

and set

$$W_k := Z_k - \frac{R^2}{3}.$$

Then

$$\boxed{W_1 + W_2 + W_3 = 0}.$$

Hence any one of them is determined by the other two:

$$W_3 = -(W_1 + W_2).$$

What matters here is that there is no need to presuppose a “triangle.”

First there are three complex relations W_1, W_2, W_3 , and their sum happens to be zero. When this algebraic condition is drawn in the complex plane, it can be read as a closed triangle.

That is, the logical order is not

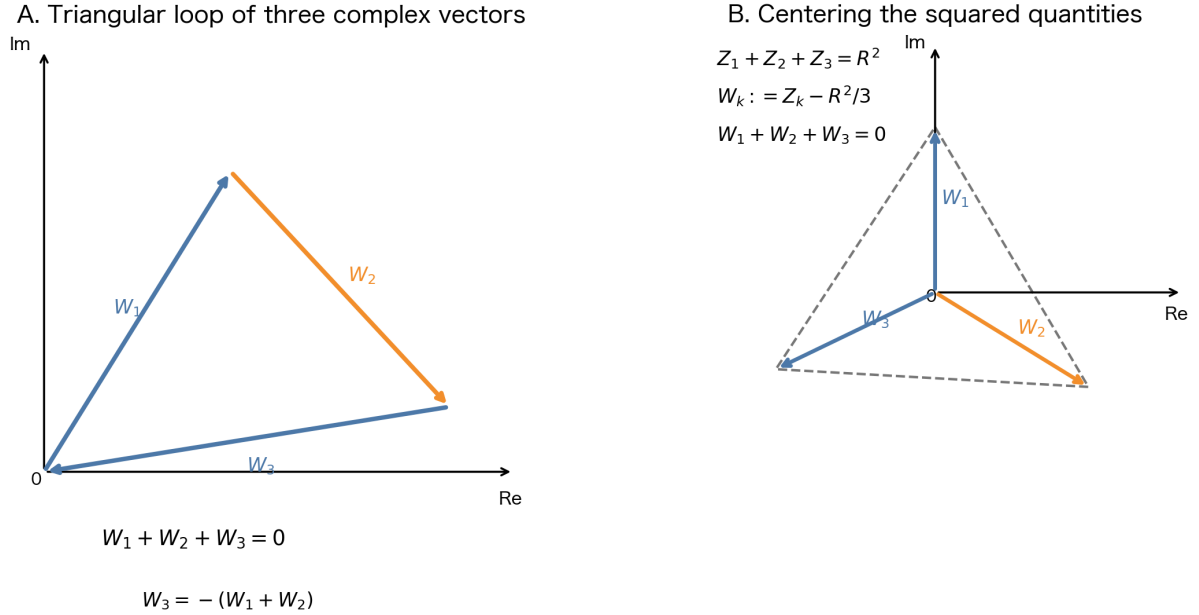
$$\boxed{\text{assume a triangle} \Rightarrow \text{loop condition}}$$

but

$$W_1 + W_2 + W_3 = 0 \Rightarrow \text{readable as a triangular loop}.$$

This ordering becomes important in the later discussion of namelessness.

Fig. 2 Three bodies: one is determined by the other two



Three bodies form the smallest nontrivial system in which a loop of relations first appears.

Figure 2: Fig. 2 Three bodies: one is determined by the other two

Fig. 2 The triangular loop of three bodies. From the zero sum of three complex relations, any one of them is fixed as the reverse of the sum of the other two, and the closed triangle is read afterwards.

7. What Four Bodies Expose: Bodies and Relations Are Not the Same

With three bodies, the number of vertices (3) coincides with the number of edges (3), so it is easy to confuse “bodies” with “relations.”

With four bodies, however, if one assumes all pairwise relations, the number of relations is

$$M = \binom{4}{2} = 6.$$

Here one is tempted to declare “four bodies, therefore the complete graph K_4 .” But this is not allowed by the methodology of this paper, because complete connectivity has not been derived from the closure of complex numbers.

In fact, even in a five-relation system consisting of the four outer edges and one diagonal on four vertices, there exist nontrivial solutions in which two triangular loops close simultaneously.

For example, with oriented relations,

$$W_{12} + W_{23} - W_{13} = 0$$

$$W_{13} + W_{34} + W_{41} = 0.$$

From the first equation,

$$W_{13} = W_{12} + W_{23},$$

and substituting into the second,

$$W_{12} + W_{23} + W_{34} + W_{41} = 0.$$

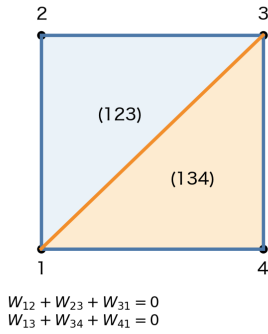
Therefore,

$$\boxed{\text{triangular loops close} \not\Rightarrow \text{complete connectivity}}.$$

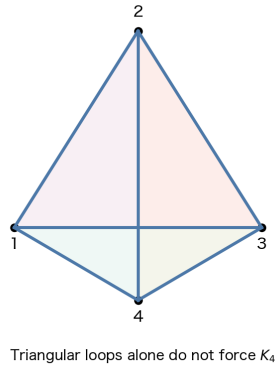
By this counterexample, the complete-network hypothesis is excluded from the thought experiment.

Fig. 3 Four vertices: triangular loops alone do not force the complete network

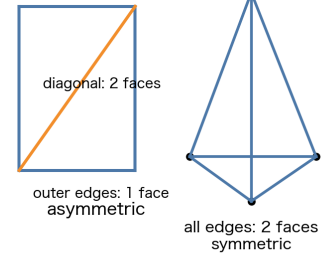
A. Four vertices, five relations



B. Complete network K_4



C. Faces shared by edges



Closing triangular loops and a fully symmetric relational structure preserving namelessness are separate issues.

Figure 3: Fig. 3 Four-vertex network comparison of triangular loops

Fig. 3 On four vertices, the complete network K_4 cannot be derived from the mere existence of triangular loops. Even with 4 vertices and 5 relations two triangular loops can close, so complete connectivity is an additional assumption.

8. A Dual Turn: From Vertex-Centered to Edge-Centered

To read structure without first laying down a complete graph, we invert the viewpoint.

Conventional pictures tend to think

$$\text{vertices} \rightarrow \text{edges} \rightarrow \text{faces}.$$

In this paper, however, the complex relations Z_e are basic, so we read instead

$$\boxed{\text{relations (edges)} \rightarrow \text{loops} \rightarrow \text{faces}}.$$

If three relations Z_1, Z_2, Z_3 satisfy

$$Z_1 + Z_2 + Z_3 = 0,$$

then those three relations can be read as one closed face.

Here we do not specify in advance “which vertices are shared.” The loop structure comes first; vertices and cell structure are read afterwards, dually.

With this ordering, there is no need to postulate complete networks or vertex names as primitive structure.

9. Loop Residual

When a loop does not close completely, we write

$$\boxed{\sum_{e \in \partial f} \epsilon_{fe} Z_e = K_f}, \quad K_f \in \mathbb{C}.$$

$\epsilon_{fe} = \pm 1$ is the sign relative to the orientation of the loop.

If $K_f = 0$ the loop closes completely; if $K_f \neq 0$ it carries a complex residual.

What matters is that K_f is again an ordinary complex number:

$$\boxed{Z_e \in \mathbb{C} \Rightarrow K_f \in \mathbb{C}}.$$

Hence “edge quantities” and “face residuals” do not change the algebraic type.

10. The Residual as a Central-Projection Radius

Since $K_f \in \mathbb{C}$, there exists $C_f \in \mathbb{C}$ such that

$$\boxed{K_f = C_f^2}.$$

Therefore

$$\sum_{e \in \partial f} \epsilon_{fe} Z_e = C_f^2$$

can be read as a complex version of a central-projection radius written in squared quantities.

At this stage there is no need to add “curvature” as an entirely separate variable. Complete closure means $C_f = 0$; non-closure means a finite complex radius.

When one reads this as a central-projection constant-curvature geometry, for non-closed faces ($K_f \neq 0$) the curvature scale is the inverse squared radius:

$$\kappa_f \propto \frac{1}{C_f^2} = \frac{1}{K_f}.$$

This curvature readout is a correspondence map. In particular, restricting to the real positive branch

$$K_f = C_f^2 \in \mathbb{R}_{>0}$$

and adopting the standard convention for the radius of curvature, one recovers the real constant-curvature readout

$$\kappa_f = \frac{1}{K_f}.$$

The geometric meaning carried by a general complex K_f , in particular $\text{Im } K_f \neq 0$, is not derived in this paper and is left as a connection problem.

What matters in this paper is that K_f was not added from outside as a “curvature variable”: the loop residual itself already exists as the quantity determining the central-projection radius.

11. Cancellation of Shared Edges and Face Flux

Consider gluing two triangular faces along one internal edge.

Let the residuals of faces f_1 and f_2 be

$$K_{f_1} = Z_{12} + Z_{23} - Z_{13}$$

$$K_{f_2} = Z_{13} + Z_{34} + Z_{41}.$$

Adding the two, the shared edge Z_{13} cancels:

$$-Z_{13} + Z_{13} = 0,$$

leaving

$$\boxed{K_{f_1} + K_{f_2} = Z_{12} + Z_{23} + Z_{34} + Z_{41}}.$$

That is, internal relations disappear and only the relations touching the external boundary remain.

Reading K_f as a face flux, we obtain the discrete Stokes-type structure

$$\boxed{\text{cancellation of shared internal relations} \Rightarrow \text{only the external boundary flux remains}}.$$

Here too, Stokes’ theorem was not placed at the starting point. This is the result of following the algebraic fact that when oriented loops are glued, the shared term appears with opposite signs.

12. Reading a Closed Polyhedron as a Condensate

Consider a structure $\mathcal{C}$ in which multiple faces are glued together and the outer shell is closed.

Each face carries a complex flux K_f , and on a face shared by adjacent cells the orientations are opposite, so the contribution of an internal face cancels:

$$K_f + (-K_f) = 0.$$

Therefore, in the flux readout defined in this paper, shared internal relations cancel and the quantity remaining on the boundary is read as the effective quantity.

However, **it is not proven in this paper that this boundary flux exhausts all irreducible relational information of the many-body system.** The possibility that higher-order relational quantities exist independently, not captured by pairwise relations and loop residuals, is explicitly left as an open problem in v1.2.

This paper proposes to call this closed polyhedral relational structure a **condensate**:

$$\boxed{\text{condensate} = \text{a closed relational complex whose internal relations cancel on shared faces}}.$$

This is not a definition of “particle = point.” A condensate is defined as many complex relations forming a closed face structure that can be read from outside as a single effective object.

This interpretation is a physical readout, not a mathematical identity, and we keep it distinguished as an axiom candidate / interpretation candidate.

13. Local Flux and Global Flux

Consider the closed polyhedral condensate of this paper, that is, a closed, oriented, 2-manifold-like face complex $\mathcal{C}$ in which every edge belongs to exactly two faces with opposite orientations. Let $F(\mathcal{C})$ denote the set of oriented faces constituting the condensate $\mathcal{C}$. With the face flux

$$K_f = \sum_{e \in \partial f} \epsilon_{fe} Z_e,$$

we have

$$\sum_{f \in F(\mathcal{C})} K_f = \sum_e \left(\sum_{f \in F(\mathcal{C})} \epsilon_{fe} \right) Z_e = 0.$$

This holds because for each shared edge the incidence signs $+1$ and -1 each appear exactly once; it is a discrete identity of $\partial^2 = 0$ type. This structure corresponds to the commutative version of the structure in which reference [4] reads the closure constraint as a Bianchi identity.

Therefore, the vanishing of the global flux is not an additional condition but a consequence of this closed oriented structure. It does not, however, imply

$$K_f = 0$$

for each face. Faces with $K_f \neq 0$ can exist locally.

This is important: a structure is possible in which nonzero interactions persist inside while the whole remains closed.

Therefore,

$$\boxed{\text{global flux zero} \neq \text{local flux zero}}.$$

If one adopts a physical correspondence to gauge theory, the nonzero fluxes on internal shared faces can be read as modes of internal gauge exchange. In QED-like vocabulary an interpretation close to “exchange of virtual photons” is possible, but this is not a particle identification derived directly from the algebra of this paper.

14. Hierarchy from the Flux Being Complex

An important feature of this construction is that the face flux K_f is itself a complex number:

$$K_f \in \mathbb{C}.$$

Hence the external flux of a condensate can be treated as a new relational quantity at the next level.

Schematically,

$$Z_e \in \mathbb{C} \longrightarrow K_f \in \mathbb{C} \longrightarrow K_c \in \mathbb{C} \longrightarrow \dots$$

That is, without changing the type

$$\boxed{\mathbb{C} \rightarrow \mathbb{C} \rightarrow \mathbb{C}},$$

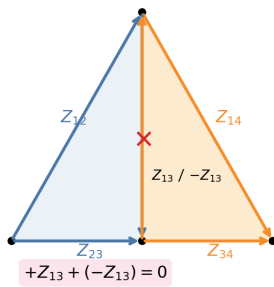
we can read recursively

$$\text{relations} \rightarrow \text{faces} \rightarrow \text{condensates} \rightarrow \text{relations between condensates}.$$

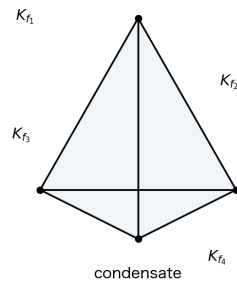
This property allows small closed structures to be glued into larger closed structures — hierarchical condensation.

Fig. 4 Internal cancellation and flux: reading a closed polyhedron as a condensate

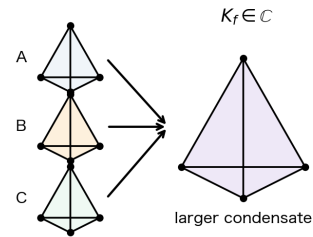
A. Two adjacent triangular faces



B. Face set of a closed polyhedron



C. Hierarchical growth of condensates



Shared internal relations cancel; from outside, the closed polyhedron appears as a single condensate.

Figure 4: Fig. 4 Internal cancellation and the hierarchical growth of condensates

Fig. 4 Oppositely oriented contributions of shared relations cancel internally, and a closed set of faces is read from outside as a single condensate. Since the face flux $K_f \in \mathbb{C}$ can be reused as a relation at the next level, the hierarchy stays within the same complex-number type.

15. Complete Closure and the Indistinguishability of “Universe” and “Grand-child Universe”

Suppose the relational flux between a closed condensate and its exterior has vanished completely.

More strongly than a mere zero total: if no boundary relation readable as an external connection exists, or if the entire effective flux on the external boundary has vanished, the system is relationally cut off from the outside.

For an internal observer, there is then no way to distinguish, without external relations, whether the system is “the whole universe” or “a grandchild universe generated inside a larger system.”

Therefore the readout

a completely closed relational world carries no absolute label of “the whole universe”

becomes possible.

This is a physical hypothesis; it is not a derivation of actual existence from the complex-number algebra alone.

16. Composition from Past to Present and the Direction of Time

The same complex-sum structure can also be used for a temporal readout.

Let the squared quantities fixed in the past be

$$Z_1, \dots, Z_n.$$

Then the present composite quantity

$$K_n = \sum_{j=1}^n Z_j$$

is uniquely computable.

After the next relation Z_{n+1} is realized,

$$K_{n+1} = K_n + Z_{n+1}.$$

In this sense, no matter how many history items there are, the update contracts to the two-body composition

$$(K_n, Z_{n+1}) \rightarrow K_{n+1}.$$

On the other hand, there is no rule that uniquely factorizes or generates the future Z_{n+1} from the present K_n alone. Therefore,

$$\boxed{\text{past} \rightarrow \text{present}}$$

is definable as composition, whereas

$$\boxed{\text{present} \rightarrow \text{future}}$$

requires an additional generation rule.

This paper does not present this as a proof of the arrow of time itself, but retains it as the **asymmetry between the composition direction and the generation direction**.

17. An Auxiliary Relation to the Log Readout

For the multiplicative relation of nonzero complex numbers z_1, z_2 ,

$$z_3 = z_1 z_2,$$

we have $z_3 \neq 0$, and with properly managed branches the complex logarithm gives

$$\text{Log } z_3 = \text{Log } z_1 + \text{Log } z_2.$$

For the self-product

$$z \mapsto z^2,$$

take $z \neq 0$ and, on the chosen logarithmic branch, set

$$q := \text{Log } z.$$

Then in Log space

$$q \mapsto 2q,$$

which reads as additive self-doubling.

On the other hand, to read the sum of squared quantities

$$\sum_n Z_n = K$$

in Log space, set for each nonzero Z_n

$$q_n := \text{Log } Z_n.$$

Zero terms contribute nothing to the sum and can be excluded. Only when $K \neq 0$,

$$\text{Log } K = \text{Log} \left(\sum_n e^{q_n} \right).$$

For $K = 0$, $\text{Log } K$ is undefined.

This Log readout is not essential to the loop/flux construction of this paper, but it can be retained as an auxiliary readout connecting the multiplicative foundation with additive observational representations.

18. History of Deleted Hidden Assumptions

If only the conclusions of this paper were presented, it would look like existing discrete gauge theory rewritten in complex numbers. We therefore state explicitly the assumptions actually removed during the thought experiment.

Candidate	Initial reading	Verdict	Reason
Complete network K_N	every relation connects to every relation	deleted	not derivable from triangular loops alone; a 4-vertex, 5-edge system with one diagonal also has loop solutions
Ratio z_2/z_1	minimal two-body relation	removed from the foundation	easily smuggles in a comparison standard at readout
Equality test $z_1 = z_2$	primitive same/different test	removed from the foundation	no need to place an equality detector as a prior physical operation
$\sum z_n^2 = 0$ for two bodies	condition for two waves	deleted	zero closure is not imposed as an extra condition in this thought experiment
Triangle	a priori geometry of three bodies	deleted	readable as a loop after $W_1 + W_2 + W_3 = 0$
Vertex sharing	connection principle of edges	on hold	vertices are not presupposed; we moved to the dual edge-to-face readout
Adding a curvature variable	a field separate from the loop residual	unnecessary	$K_f = C_f^2$ reads as a complex central-projection radius
Adding gauge particles	separate particles carrying flux	demoted to physical interpretation	local flux propagation may be read as gauge exchange, but independent particles are not derived
Completeness of many-body information	Z_e and K_f exhaust all relational information of a many-body system	on hold	implicitly assumed in v1.1; noticed to be underived on the occasion of Anegawa et al. [6]. It has not been shown that irreducible higher-order relations do not exist independently

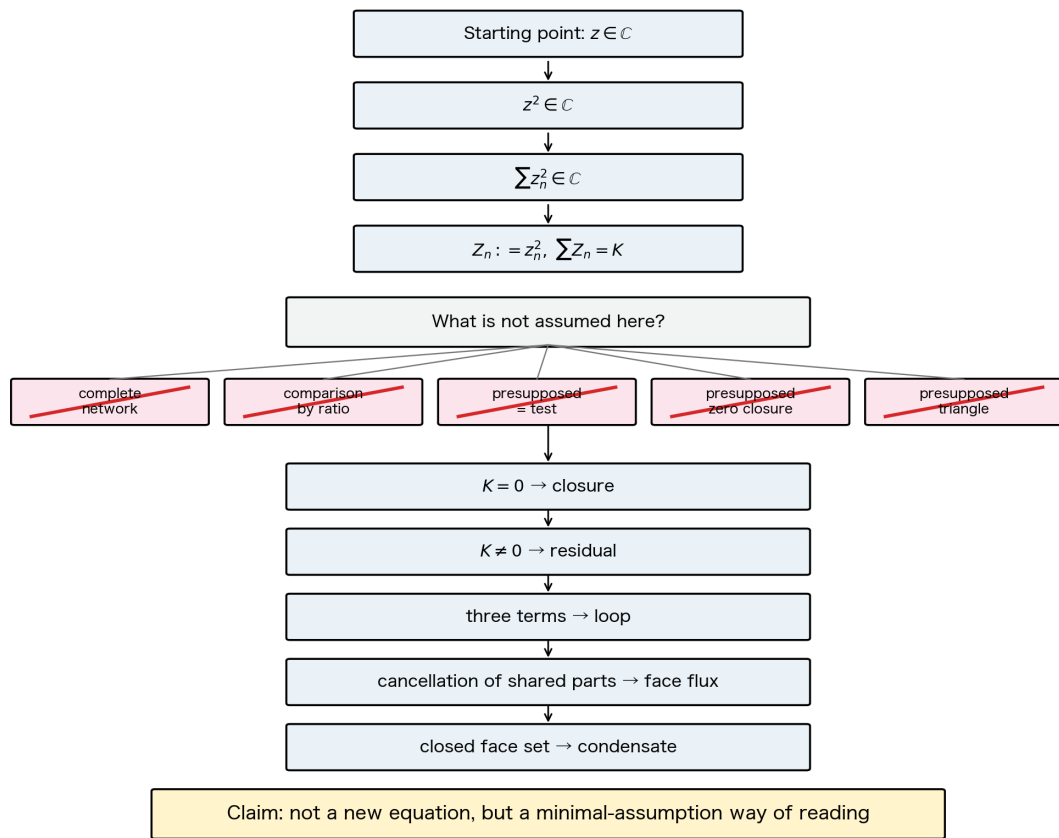
As the table shows, this paper moved in the direction of deriving or deleting assumption candidates rather than adding assumptions.

Fig. 5 The method of the thought experiment. Starting from the closure of complex numbers, at each stage we check whether a complete network, comparison by ratio, equality tests, zero closure, or presupposed triangles have been assumed, shave off the unnecessary assumptions, and adopt only the readouts that remain.

19. Axiom Candidates Remaining at This Point

Apart from the algebra of complex numbers itself, the following candidates remain at this stage.

Fig. 5 Method of the thought experiment: what remains after shaving off assumptions



The core lies not in what was introduced, but in what never needed to be introduced.

Figure 5: Fig. 5 Method flowchart of the thought experiment

19.1 Simultaneous existence of multiple complex relations

It cannot yet be said that “multiple occurrences” have been generated from a single value $z \in \mathbb{C}$ by algebra alone. Therefore the existence, or a generation rule, of multiple relations is an axiom candidate.

19.2 Generation of orientation and incidence

The structure in which a shared edge carries opposite signs in two faces follows automatically once oriented boundaries are defined. But whether the oriented incidence structure itself necessarily emerges from nameless complex relations is unsettled.

19.3 Reading a loop as a face

Drawing the zero sum of three complex relations as a closed triangle is mathematically natural, but identifying it with a physical “face” is a readout rule.

19.4 Reading a closed face complex as a condensate

Reading an internally cancelled closed polyhedron as one effective object is the main physical interpretation candidate of this paper.

19.5 Completeness of many-body information via pair relations and loop residuals

Although not stated explicitly in v1.1, parts of the argument implicitly used the assumption that

$$\mathcal{I}_{\text{pair+loop}} := \{Z_e, K_f\}$$

suffices to represent the relational information of a many-body system. However, the completeness

$$\boxed{\{Z_e, K_f\} \stackrel{?}{\Rightarrow} \mathcal{R}_{\text{full}}}$$

is unproven. In particular, whether irreducible higher-order relational quantities such as

$$R_{123}, \quad R_{1234}, \quad \dots$$

are independently necessary, or whether they can be derived from Z_e and K_f , is unsettled. This paper does not adopt this point as a new axiom but **keeps it on hold as an axiom candidate / derivation problem**.

These are the objects of future study: whether they can be derived further from complex-number algebra and namelessness.

20. Connection to Existing Gauge Theories

Comparing with existing theories after obtaining the construction above, the structural similarity with lattice gauge theory becomes clear.

In Wilson’s lattice gauge theory, link variables on a discrete lattice and closed loops play the central role [1]. In Hamiltonian lattice gauge theory as well, electric flux and link structure are fundamental [2].

This paper does not put a gauge group or a lattice first, but reading

$$K_f = \sum_{e \in \partial f} \epsilon_{fe} Z_e$$

as a face flux, the format “quantities on edges $\rightarrow$ loops $\rightarrow$ residuals on faces” is close to commutative lattice gauge theory.

The difference lies in the starting point. In this paper:

1. No lattice is assumed.
2. No complete graph is assumed.
3. No vertices are assumed first.
4. No specific gauge group is assumed first.
5. Both squared quantities and face residuals belong to the same $\mathbb{C}$.

Hence this is not a simplified construction of existing gauge theory; the order is that gauge-theoretic structure was connected only after following the closure of complex numbers.

21. Comparison with Closure Defects, the Bianchi Identity, Bubble Networks, and Multi-partite Entanglement

Livine relaxes the closure constraint in the coarse-graining of spin networks and interprets the closure defect as a local measure of curvature inside the coarse-grained region [3].

Charles and Livine reinterpret the closure constraint of the hyperbolic tetrahedron as a Bianchi identity, connecting face holonomies with polyhedral closure [4].

The bubble networks of Freidel and Livine glue three-dimensional geometric cells along interfaces and handle the structure of boundary faces, fluxes, and gluing [5].

These have strong similarity to this paper's

$$\text{relations} \rightarrow \text{loops} \rightarrow \text{residuals} \rightarrow \text{face flux} \rightarrow \text{closed cells}.$$

The differences are equally clear. In the prior work, structures such as spin networks, $SU(2)$ holonomies, spinors, graphs, and cell complexes exist from the start. This paper does not take them as the starting point; it examines how far one can reach the same type of readout from

$$z \in \mathbb{C}$$

and

$$Z = z^2 \in \mathbb{C}$$

together with the closure of complex addition.

The relation to prior work is therefore positioned not as “imitation of the same structure” but as **verification of the possibility of having reached the same kind of structure independently from fewer premises**.

Anegawa, Miyata, Suzuki, Tamaoka [6] showed that in a fixed tripartition of the two-sided BTZ black hole, a quantity measuring genuine multi-partite entanglement keeps growing even after the ordinary entanglement entropies of the subsystems saturate, continuing to track the growth of the Einstein–Rosen bridge.

This result cannot be identified with the K_f or the condensates of this paper, nor does it directly support or refute the present complex-relation model. What matters for this paper is that it supplies **a concrete example that low-order readouts need not exhaust the total information of a many-body system**. Through it we recognized the implicit assumption remaining in v1.1 — “ Z_e and K_f suffice to exhaust the effective relational information of a many-body system” — and in v1.2 returned that completeness to the list of open problems.

22. Candidate Novelty

We divide the candidate novelty of this paper into mathematical novelty and interpretive novelty.

22.1 Mathematical novelty

No novelty is claimed for the closure of complex numbers, sums of squares, closed polygons, or boundary cancellation themselves.

22.2 Interpretive novelty candidates

What this paper proposes is the following readout chain:

$$\mathbb{C} \rightarrow z^2 \in \mathbb{C} \rightarrow \sum Z_n = K \in \mathbb{C} \rightarrow \text{closure / defect} \rightarrow \text{face flux} \rightarrow \text{closed condensate} \rightarrow \text{hierarchical condensate}$$

Moreover, in obtaining this chain we assumed no complete network, no fixed vertices, no fixed dimension, and no external comparison standard.

22.3 Methodological novelty candidate

In this paper, instead of adding axioms while building a model, we delete a hidden assumption each time one is discovered.

What matters in the end is therefore

not what was introduced, but what never had to be introduced.

23. Limitations and Open Problems

This paper alone does not derive the following.

1. How multiple independent complex relations are generated from a single complex number.
2. Whether the oriented incidence structure necessarily arises from namelessness.
3. Which loops are selected as actually existing faces.
4. Whether triangular loops suffice, or higher polygonal loops are independently necessary.
5. The time-evolution law of the flux.
6. How flux propagation maps to the individual gauge particles of the Standard Model.
7. Stability conditions of condensates and the readout of mass, energy, and spin.
8. Whether completely closed systems correspond to actually existing independent universes.
9. The geometric meaning of complex K_f , in particular of its imaginary part, and the correspondence from the complex central-projection radius to general non-constant curvature fields.
10. Whether the set of pairwise relations Z_e and loop residuals K_f suffices to determine the full relational information of a many-body system. This informational sufficiency can be tested by fixing all face residuals and the external boundary readout on the face-edge incidence matrix and computing, by a rank computation, the dimension of the solution space admitted for the internal Z_e .
11. Whether irreducible relational invariants of three or more bodies, $R_{123}, R_{1234}, \dots$, exist independently, underivable from pairwise and loop quantities.

Items 10 and 11 in particular make explicit, in v1.2, the hidden assumption that remained in v1.1. Their validity is currently unsettled and is left to future theoretical analysis and numerical experiments.

These are not negative evidence but connection problems for the next stage.

24. Conclusion

The starting point of this paper is one ordinary complex number:

$$z \in \mathbb{C}.$$

A complex number squared is still a complex number:

$$Z_n := z_n^2 \in \mathbb{C}.$$

Therefore

$$\sum_n z_n^2 = C^2$$

is, once squared quantities are read as basic,

$$\boxed{\sum_n Z_n = K},$$

an ordinary sum of complex numbers.

From this simple rereading, one reads a zero sum of three relations as a loop, a nonzero sum as a loop residual, and by gluing oriented loops obtains the cancellation of shared internal relations and the external boundary flux. Since the flux is again complex, it can be reused as a relation at the next level. A closed set of faces can be read as one effective object with its internal degrees of freedom cancelled — a condensate.

What matters is that in obtaining these we did not first posit complete networks, fixed vertices, triangles, gauge groups, or external scales. Rather, by doubting them as assumption candidates and deleting those that could not be derived, a loop/flux structure similar to existing gauge theory appeared afterwards.

The central claim of this paper is therefore not a new equation:

Give the trivial closure of complex numbers a relational readout with minimized assumptions, and the hierarchy of loops

However, this unified readout does **not claim to exhaust the full relational information of many-body systems**. In v1.2 we withdrew the implicit assumption that the pairwise relations Z_e and the loop residuals K_f suffice, and left as an open problem whether irreducible higher-order relations are independently necessary.

We do not claim that this framework is a final physical theory. But as a minimal basis for clarifying which additional assumptions are truly indispensable, it merits examination.

References

This paper uses no self-citation. The following are external references consulted, after the construction of this paper was obtained, to check structural similarities with existing theories.

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Appendix A: The Minimal Series of Formulas

Extracting only the algebraic skeleton of this paper gives the following series.

$$z_n \in \mathbb{C}$$

$$Z_n := z_n^2 \in \mathbb{C}$$

$$K := \sum_n Z_n \in \mathbb{C}$$

$$K = C^2, \quad C \in \mathbb{C}$$

Loop:

$$\sum_{e \in \partial f} \epsilon_{fe} Z_e = 0$$

Loop residual:

$$\sum_{e \in \partial f} \epsilon_{fe} Z_e = K_f$$

Central-projection-radius readout:

$$K_f = C_f^2$$

Shared-edge cancellation:

$$(+Z_e) + (-Z_e) = 0$$

Composition of multiple faces:

$$\sum_f K_f = \sum_{e \in \partial \Omega} \epsilon_e Z_e$$

Global identity on a closed oriented face complex (derived identity structure):

$$\boxed{\sum_{f \in F(\mathcal{C})} K_f = 0}$$

$F(\mathcal{C})$ is the set of oriented faces constituting the condensate $\mathcal{C}$. On a closed polyhedral boundary in which every edge belongs to two faces with opposite orientations, this holds identically by incidence cancellation, i.e., by the discrete $\partial^2 = 0$.

Hierarchization:

$$K_f \in \mathbb{C} \quad \Rightarrow \quad K_f \text{ is reusable as a relational quantity at the next level}$$

Appendix B: Ledger of Axiom Candidates

Object	Current classification	Reason
$z \in \mathbb{C}$	mathematical background	complex numbers adopted as the descriptive language
$z^2 \in \mathbb{C}$	derived identity structure	closure of complex numbers
$\sum z_n^2 = C^2$	derived identity structure	$\sum z_n^2 \in \mathbb{C}$ and existence of square roots
Reading $Z_n = z_n^2$ as the basic quantity	readout definition	entry point of the new physical interpretation
Existence of multiple relations	axiom candidate	multiple occurrences not yet derived from a single complex number
Reading a loop as a face	readout candidate	natural as a geometric representation, but physical identity unproven
Oriented incidence	axiom candidate / derivation problem	needed for shared-edge cancellation; derivation from namelessness incomplete
Reading $K_f = C_f^2$ as a complex central-projection radius	geometric readout	re-expression of the same sum-of-squares structure
Closed polyhedron = condensate	physical interpretation candidate	has internal cancellation and external effectivization
Local flux = gauge exchange	connection problem with existing theory	similar to lattice gauge theory, but particle identity underived
Completeness of many-body information $\{Z_e, K_f\} \Rightarrow \mathcal{R}_{\text{full}}$	axiom candidate / derivation problem	implicitly assumed in v1.1 but unproven; the possibility of independently existing irreducible higher-order relational quantities cannot be excluded
Complete network	deleted	not derivable from loop conditions alone